

Cristian Barrera

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SUMMARY

Data Scientist focused on medical imaging and machine learning, with 7+ years across academic research, biotech, and production clinical pipelines. Strong experience in OCT, deep learning, distributed training, and MLOps, with a track record of shipping practical tools used by research and clinical teams.

EDUCATION

Georgia Tech & Emory University, Atlanta, GA

Aug 2020 – Jan 2026

Ph.D., Biomedical Engineering

Medical Imaging, Histopathology, Ophthalmology

National University of Colombia

Jan 2017 – Dec 2019

Master of Engineering, Biomedical Engineering

Medical Image Processing, Machine Learning, Genomics

South Colombian University

Jan 2011 – Dec 2015

Bachelor of Engineering, Electronics

Digital Electronics, Signal Processing, Telecommunications

EXPERIENCE

Data Scientist III

Nov 2024 – Present

Cleveland Clinic (Cole Eye Institute) - Cleveland, OH

- Lead OCT foundation model development by pretraining iVisionFM on 3.2M B-scans, then fine-tuning on hundreds of thousands to millions of scans for layer-specific tasks (ILM, BM, RPE, and related boundaries).
- Designed a mixture-of-experts architecture with a pathology-aware router, delivering 30–40% performance improvement versus a TransUNet-only baseline on complex pathology cohorts.
- Architected distributed training and inference across 20+ GPUs on 10+ Linux workstations with containerized execution.
- Built an Airflow-based orchestration workflow for OCT run submission across multiple computers, integrating 20+ containerized models.
- Delivered a React and PostgreSQL platform for run submission, tracking, and quality-control visibility at 50–100 runs per day.

Owner

2024 – Present

BARTEK LLC (bartekllc.org) - Cleveland, OH

- Founded BARTEK LLC and delivered a 9-month Veterans Affairs contract (Sep 2025 – Jun 2026) to build a histopathology image processing platform.
- Built a web workflow for users to submit histopathology runs, monitor progress, and review outputs.
- Developed a containerized multi-model pipeline with foundational models and run-allocation logic across compute resources.
- Implemented an AI assistant to recommend the best models to execute for each submission context.

Graduate Research Scientist

Feb 2018 – Jan 2024

Emory University - Atlanta, GA

- Designed data models and visualizations for 4,000+ histopathology images for cancer and biomarker research.
- Developed machine learning models for patient outcomes and immunotherapy response contributing to peer-reviewed publications.

- Built scalable deep learning and traditional ML pipelines for biomedical imaging and clinical datasets.
- Collaborated with multidisciplinary teams to translate computational results into clinically relevant findings.

Imaging Scientist Intern

Jun 2023 – Aug 2023

Genentech - South San Francisco, CA

- Improved data processing efficiency by 30% through Python-based workflows.
- Developed ETL patterns for immunofluorescence imaging to integrate heterogeneous research datasets.
- Partnered with scientists to deliver reproducible analysis outputs for imaging studies.

Visiting Scientist

Nov 2017 – Jan 2020

Case Western Reserve University - Cleveland, OH

- Supported imaging and data analysis workflows in biomedical engineering research collaborations.
- Applied deep learning and signal processing methods to experimental datasets.
- Delivered reproducible code and documentation for faculty and lab teams.

Biomedical Engineering Contractor

Jan 2017 – Dec 2019

National University of Colombia - Bogota, Colombia

- Developed deep learning face recognition tools for longitudinal child tracking programs.
- Designed data collection and evaluation workflows to monitor accuracy and false alerts in deployment.
- Supported surface EMG testing for maternal-fetal heart rate capture and downstream signal analysis.

PROJECTS & CURRENT WORK

OCT Foundation Model and MoE Segmentation Stack. Pretrained iVisionFM on 3.2M OCT B-scans and fine-tuned for retinal layer segmentation (including ILM, BM, and RPE) with hundreds of thousands to millions of labeled examples, then added a pathology-aware mixture-of-experts router to improve complex-case performance by 30–40% versus a TransUNet-only approach.

Ophthalmology OCT Workflow Platform. Distributed Airflow and container-based workflow for OCT processing across multiple computers, integrating 20+ models behind a React interface for run submission, monitoring, and quality control, with PostgreSQL tracking 50–100 daily runs.

SKILLS

Main Languages: Python, R

Frameworks/Tools: PyTorch, TensorFlow, Airflow, React, Flask, Docker, Podman, Redis, PostgreSQL, MLflow, Git, Linux, HPC, AWS, Azure

Domains: Medical Imaging, OCT, Histopathology, Machine Learning, Deep Learning, Data Engineering, MLOps, Ophthalmology